

Sahrdaya College of Engineering and Technology, Kodakara		
Answer Key		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY		
SECOND SEMESTER B.TECH DEGREE EXAMINATION, JUNE 2023		
Course Code: EST200		
Course Name: Desgin and Engineering		
PART A Answer all questions, each carries 4 marks		
Q1	How is engineering design different from other kinds of design?	
ans	An engineer is a creative, ingenious person and Engineers create ingenious solutions to societal problems. Engineering design process can be defined in many ways as follows. Engineering design is a systematic, intelligent process in which engineers generate, evaluate, and specify solutions for devices, systems, or processes whose form(s) and function(s) achieve clients' objectives and users' needs while satisfying a specified set of constraints. In other words, engineering design is a thoughtful process for generating plans or schemes for devices, systems, or processes that attain given objectives while adhering to specified constraints. The engineering design process is a methodical series of steps that engineers use in creating functional products and processes. It is a decision-making process, in which the basic sciences, mathematics, and engineering sciences are applied to optimally convert resources to meet a stated objective. Among the fundamental elements of the design process is the establishment of objectives and criteria, synthesis, analysis, construction, testing, and evaluation.	
Q2	Explain how to identify customer requirements of design.	
	1.12.1 Need identification The design process starts with need identification. At times the user, the client and the customer could be the same or different. To start with, the Designer has to meet the needs of the Client as well as the User. The entity that has the need to use the design is the user. For any Design to be taken up, there should be a need gap or a problem that needs a solution. This gap or need could be identified by the user, an observer or by an organization. Through interviews with potential purchasers, focus groups, and by observing similar products in use, researchers identify these gaps or user needs. The need for a new design can be generated from several sources, including the following: 1. Client request – In a design company, a client may submit a request for developing an artifact. It is often unlikely that the need will be expressed clearly. The client may know only the type of product that he or she wants; for example, “I need a safe ladder.”	

	2. Modification of an existing design – Often a client asks for a modification of an existing artifact to make it simpler and easy to use. In addition, companies may want to provide customers with new, easy-to-use products. 3. Generation of a new product – In all profit-oriented industries, the attention, talent, and abilities of management, engineering, production, inspection, advertising, marketing, sales, and servicing are focused on causing the product to return profit for the company and in turn for company stockholders. For an industry to survive in today's world, it must continue to grow; it cannot afford to remain static. This growth, throughout history, has been built on new products. A product will peak out when it has saturated the market and then begin to decline. It is obvious that an industry must seek out and promote a flow of new product ideas. These new products are usually protected by applying for patents. The needs of customers outside of the company are important to the development of the product design specifications for new or improved products. A second set of critical constituents are the internal customers, such as a company's own corporate management, manufacturing personnel, the sales staff, and field service personnel must be considered. For example, the design engineer who receives information on the properties of three potential materials for his or her design is an internal customer of the materials specialist.
Q3	Explain the term design functions as applicable to engineering design.
	2.3.3 Design functions Functionality is the ability of a design to meet its product's operational criteria. The product designed must do its intended job. If the product is a chair, it has to hold your backside off the ground. If it's a table, you must be able to sit at it, and you must be able to lie in a bed. Some designs may have several functions. The main function termed the primary function and the other functions the secondary functions which may include safety, strength, ease of use, efficiency, simplicity, durability etc. Commercial product design often has a strong emphasis on functional efficiency that places aesthetics behind the qualities of easy, reliable operation. However, in the majority of consumer products, aesthetic qualities and image drive the market, often at the expense of function.
Q4	How does design thinking approach help engineers in creating innovative and efficient designs?
	Design thinking is a process for solving problems by prioritizing the consumer's needs above all else. It relies on observing, with empathy, how people interact with their environments, and employs an iterative, hands-on approach to creating innovative solutions. Design thinking is "human-centered," which means that it uses evidence of how consumers (humans) actually engage with a product or service, rather than how someone else or an organization thinks they will engage with it. To be truly human-centered, designers watch how people use a product or service and continue to refine the product or service in order to improve the consumer's experience. This is the "iterative" part of design thinking. It favors moving quickly to get prototypes out to test, rather than endless research or rumination. In contrast to traditional problem-solving, which is a linear process of identifying a problem and then brainstorming solutions, design thinking only works if it is iterative. It is less of a means to get to a

		single solution, and more of a way to continuously evolve your thinking and respond to consumer needs.
5		Briefly explain the role of basic principles of mathematical modelling in design. •MATHEMATICAL MODELS are central to design because we have to be able to predict the behavior of the devices or systems that we are designing. •It is important for us to ask: How do we create mathematical models? How do we validate such models? How do we use them? And, are there any limits on their use? •Basic Principles of Mathematical Modeling involves •Why do we need a model? •For what will we use the model? •What do we want to find with this model? •What data are we given? •What can we assume? •How should we develop this model, that is, what are the appropriate physical principles we need to apply? •What will our model predict? •Can we verify the model's predictions (i.e., are our calculations correct?) •Are the predictions valid (i.e., do our predictions conform to what we observe?) •Can we improve the model?
6		What are the factors considered in communicating designs through oral presentation? •Most design projects call for a number of both formal and informal presentations to clients, users, and technical reviewers •Because of the variety of presentations and briefings that a team may be called upon to make, it is impossible to examine each of them in detail. •However, there are key elements common to most of them. •Foremost among these needs to 1. Identify the audience 2. Outline the presentation 3. Develop appropriate supporting materials 4. Practice the presentation

7	Examine how engineering students can learn design engineering through projects.
	Learning design engineering through projects is a highly effective and practical approach for engineering students. It allows them to apply theoretical knowledge to real-world problems and develop critical skills. Here's an examination of how engineering students can benefit from project-based learning in design engineering: Application of Theoretical Knowledge: Projects provide a platform for students to apply the theoretical concepts they learn in the classroom to practical situations. This hands-on experience helps solidify their understanding of engineering principles. Problem Solving and Critical Thinking: Design engineering projects often involve solving complex, open-ended problems. Students learn to think critically, analyze issues, and develop creative solutions. Interdisciplinary Learning: Many engineering projects require collaboration with students from different engineering disciplines. This interdisciplinary approach exposes students to a wider range of skills and knowledge, preparing them for the real-world, where projects are often multidisciplinary. Project Management Skills: Students learn project management skills, including planning, scheduling, budgeting, and resource allocation. These skills are essential in engineering practice and any professional setting. Hands-On Skills:

Working on design projects allows students to develop hands-on skills, such as using tools, equipment, and software relevant to their field. These practical skills are valuable in their future careers.

Prototyping and Testing:

Students gain experience in prototyping and testing, which is crucial in engineering design. They learn how to build and test their designs to ensure they meet specifications and requirements.

Creativity and Innovation:

Project-based learning encourages creativity and innovation. Students have the freedom to explore different solutions and design variations, fostering a mindset of continuous improvement.

Real-World Constraints:

In projects, students often encounter real-world constraints like budgets, time limitations, and resource availability. This prepares them for the constraints they'll face in their professional careers.

Communication and Presentation Skills:

Students develop communication and presentation skills as they need to effectively convey their design ideas and project progress to peers, instructors, and sometimes external stakeholders.

Problem Identification:

One of the key skills students learn in project-based learning is problem identification. They develop the ability to understand the underlying issues and formulate project objectives to address them.

Feedback and Iteration:

		Students learn the importance of seeking and incorporating feedback into their designs. This iterative process is crucial in improving the quality of the final product. Industry Relevance: Many project-based learning experiences are based on real-world challenges or sponsored by industry partners. This exposes students to current industry practices and expectations. Portfolio Development: Successful project outcomes can be added to a student's portfolio, showcasing their practical skills and problem-solving abilities to potential employers. Career Readiness: Ultimately, project-based learning in design engineering equips students with the skills, knowledge, and experience necessary for their future engineering careers. To make the most of project-based learning, engineering students should actively engage, collaborate, seek guidance from mentors, and reflect on their experiences. It's an invaluable part of their education and personal and professional growth.
	8	Distinguish how aesthetics and ergonomics change engineering designs.
		Aesthetics and ergonomics are two key aspects that significantly influence engineering designs, and they have distinct impacts on the final product or system. Here's how aesthetics and ergonomics change engineering designs and the key differences between them: Aesthetics in Engineering Design:

Visual Appeal: Aesthetics primarily focus on the visual aspects of a design. It involves making a product or system visually pleasing and attractive. Aesthetically pleasing designs can create a positive emotional response in users and improve the overall user experience.

Color, Shape, and Style: Aesthetics may involve choosing the right colors, shapes, and styles that resonate with the intended user audience. It often includes factors such as symmetry, proportion, and harmony in design elements.

Marketing and Branding: Aesthetics play a significant role in marketing and branding. A visually appealing product can attract more consumers and reinforce a brand's identity.

Customization: Aesthetic considerations may vary based on cultural, regional, or individual preferences. Customization can be important to cater to diverse aesthetic tastes.

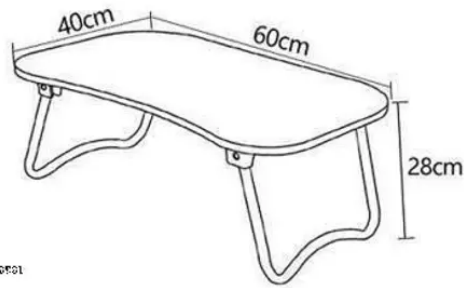
Ergonomics in Engineering Design:

User Comfort and Efficiency: Ergonomics is concerned with optimizing a design for user comfort and efficiency. It focuses on ensuring that the product or system is easy and comfortable to use, reducing the risk of strain or injury.

Human Factors: Ergonomics involves studying how humans interact with a product or system. It considers factors like body posture, reach, and range of motion to ensure that the design accommodates the user's physical and cognitive capabilities.

	Safety and Well-Being: Ergonomics aims to improve safety and well-being by minimizing the risk of user-related errors or accidents. It helps prevent repetitive stress injuries and other health-related issues. Functionality and Accessibility: Ergonomic designs prioritize the functionality of a product or system. They ensure that controls and interfaces are easily accessible and user-friendly. Key Differences: Focus: Aesthetics focus on the visual and emotional aspects of design, while ergonomics concentrate on the physical and functional aspects. User Experience: Aesthetics aim to enhance the overall user experience by creating visually pleasing designs. Ergonomics aim to improve user comfort and efficiency. Subjectivity: Aesthetics are often subjective and vary from person to person. Ergonomics is more objective and based on scientific principles to ensure usability and comfort. Interdisciplinary Nature: Ergonomics often involves collaboration between engineers, psychologists, and medical professionals, while aesthetics may involve designers, artists, and marketing experts. In practice, a successful engineering design often strikes a balance between aesthetics and ergonomics. Combining an aesthetically pleasing appearance with a user-friendly and ergonomic design can result in products and systems that are not only attractive but also functional, safe, and comfortable to use.
9	Show how designs are finalized based on the aspects of production methods.
	Formally, product design can be divided into three fundamental components:

		Appearance; Functionality; Quality.
10		Tell how ethics play a decisive role in engineering design.
		Engineering design ethics concerns issues that arise during the design of technological products, processes, systems, and services. This includes issues such as safety, sustainability, user autonomy, and privacy. Ethical concern with respect to technology has often focused on the user phase. Technologies, however, take their shape during the design phase. The engineering design process thus underlies many ethical issues in technology, even when the ethical challenge occurs in operation and use.
		PART B
		Design a study table for a child upto 5 years old specifying objectives, functions and constraints through various stages of design processes. Use hand sketches to illustrate the process.
		Problem/need To design a chair for a 5-year-old child Objectives • chair must be safe for children • easily accessible for 5 year old child • good material must be used (child safe) • must be strong • steady structure • should be able to have food while sitting Constraints (the limitations to the design) • Height of the chair must be between 30 to 40 cm • Budget must be minimum (below 500) • Material used must be strong and light weight • Safety must be priority Design 1



- ✓ Material used – Plastic
- ✓ Height of the table – 28 cm
- ✓ foldable legs are available
- ✓ Detachable
- ✓ Budget is nearly Rs 400
- ✓ Colours available – blue, red, yellow, green

Highlight of the design

The design 1 satisfies most of the objectives. The material used is safe for kids and its light weight. The plastic used material is safe for kids. The height of the chair is between the suggested height mentioned in the objective. There is a detachable food table with this chair, with no extra cost. So it can be used as a study table also. A variety of colours are available with this design.

Design 2



- ✓ Material used – wood
- ✓ Height of the chair – 40 cm
- ✓ Two closed shelves are available

	✓ Budget is – 700 rs ✓ Colour available – wood colour only Highlight of the design In this design the material used is wood. The height of the chair is 40 cm. the wood material colour only available in this design. The chair base is available with cushions for comfort In the above two designs each design meets most of the objectives/ requirements. By comparing these two designs design 1 have good plastic material used and its light weighted. Which is easy for children's use. Also its available in different colours. The design 2 is on the other hand little bit more expensive and not a good design for children. The weight of the table is more, compared with the design 1. Also the food table need to be purchased additionally. The design 1 meets more merits to the given objectives, so we can finally choose design 1 as our preferred design. Design 1
	Choose the best design for a laptop stand for avoiding overheating, also incorporate the customer requirements. Show how design objectives were finalized considering the constraints also.
	Problem/need To design a laptop stand for avoiding overheating Objectives • avoiding overheating • easy to use • god material must be used • must be strong • steady structure Constraints (the limitations to the design) • avoiding overheating • Budget must be minimum (below 500) • Material used must be strong and light weight • Safety must be priority Design 1



shutterstock

IMAGE ID: 1184117761
www.shutterstock.com

Colour Silver

Form Factor Aluminum alloy

Compatible Devices Laptops, Tablet, Books, Projector, MacBook Pro/Air, Dell XPS, HP, Samsung Laptops and Other 11-17 Inch NotebookLaptops, Tablet, Books, Projector, MacBook Pro/Air, Dell XPS, HP, Samsung Laptops and Other 11-17 Inch Notebook

Mounting Type Tabletop

Highlight of the design

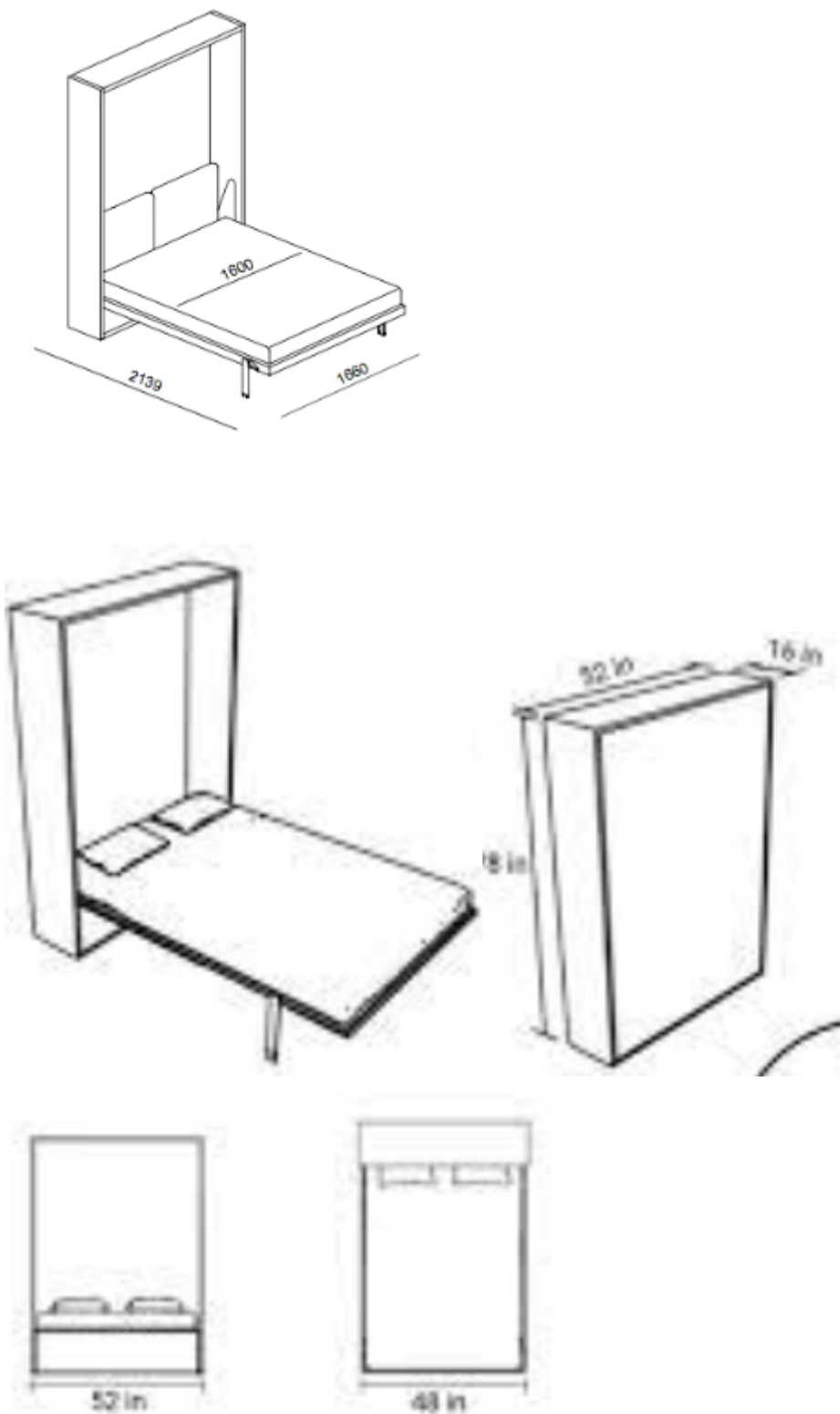
Ergonomic Design - The adjustable stand elevates your laptop by 10.6 inches to a perfect eye level and prevents you from hunching over your screen, which reduces neck and shoulder pain. Moreover, the edge of the stand is so smooth that protects you from being scratched.

Sturdy and Foldable - The office computer stand is made of thickened aluminum alloy, so the stand is sturdier and less wobbly. The Non-slip Silicone pads stick tightly and ensure your laptop stable on the stand. And the rubber feet prevent the stand from slipping on your desktop. And the lightweight laptop stand is Foldable.

Wide Compatibility - This laptop riser is compatible with all laptop models and sizes up to 17 inches, such as MacBook Pro/Air, Surface Laptop, Dell XPS, HP, Samsung, Lenovo, Google Pixelbook, Acer, Chromebook and more.

Ventilation Design - EPN laptop riser is made of aluminum alloy material, sturdy, rust-proof; scratch-proof, the heat conduction alloy and ventilation hollow design, work together to cool down your computer and prevent overheating.

	Multi-Use & Practical - Multi-use on many occasions such as office, meeting room, library, coffee shop, kitchen and music playing house to hold the laptops, tablets, projectors, menu, music score, magazines etc. An excellent useful gift for your friends and family.Design 2  Colour Dark Grey Form Factor Table Stand Compatible Devices Laptops Mounting TypeTabletop Highlight of the design Made of Aluminum Alloy with Anti Slip Silicon Rubber pads to keep laptop in place Seven Adjustable Levels It Supports Upto 5kg Weight Premium and Stylish Design Foldable Stand for Laptops up to 17 inches (Including Dell, HP, Lenovo, Macbook, Mi, Surface, Thinkpad, Asus, etc) Portable and Easy to Carry with the Pouch Open heat dissipation - helps in Preventing Overheating In the above two designs each design meets most of the objectives/ requirements. By comparing these two designs design 1 is adopted
	Construct a number of possible designs and then refine them to narrow down to the best design for a baby seater in a car. Show how convergent-divergent thinking helps in the process. Illustrate with hand sketches.

		Design a carry bag for shopping purposes. Illustrate the various stages involved in design thinking. Sketch the final design.
		Module 3
		Draw the detailed drawing of a foldable bed with design detailing, scale drawings and dimensions. Use hand sketches. 

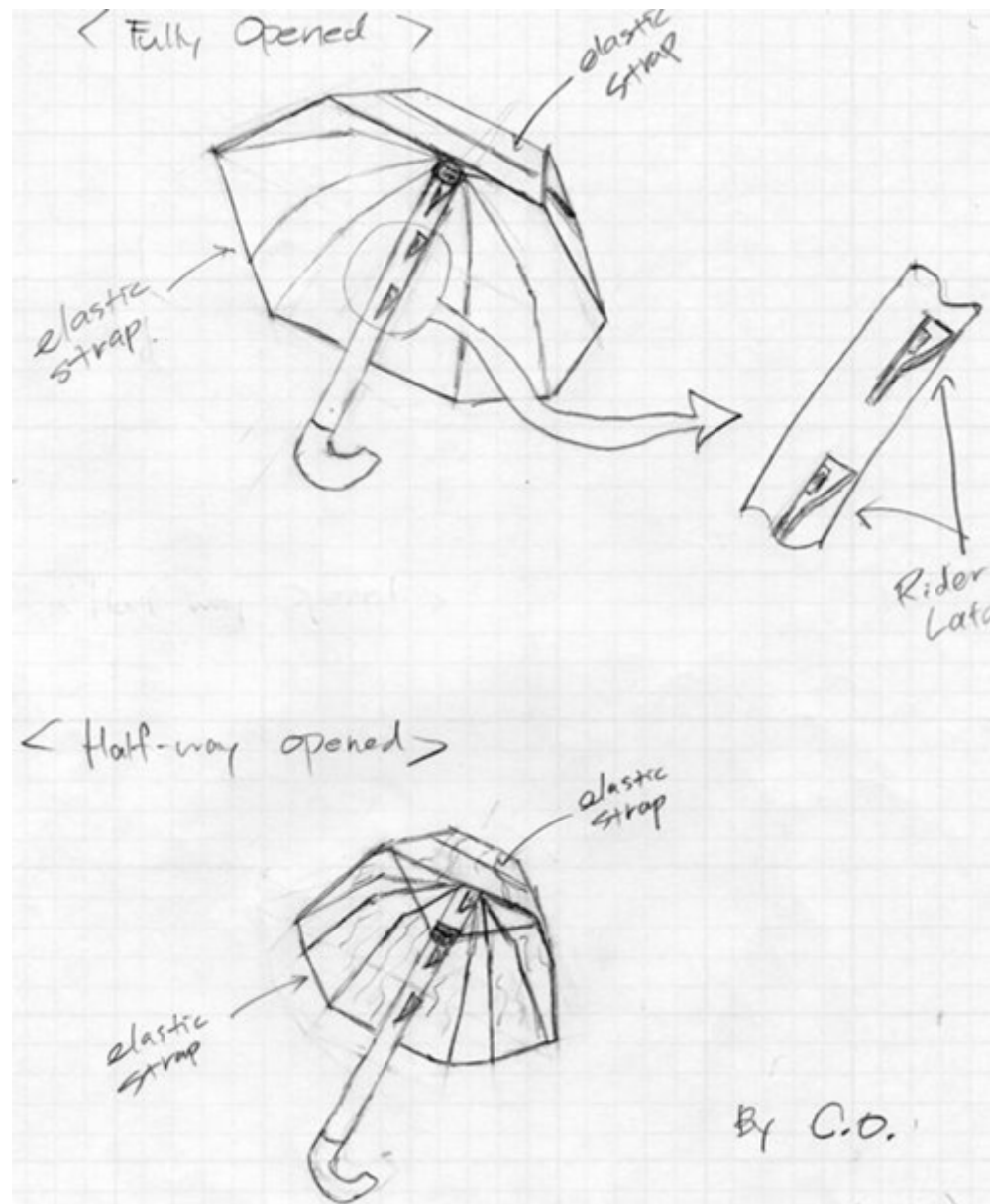
		Explain the steps for oral presentation for marketing your own product within limited budget? Illustrate with an example and suitable figures. •Most design projects call for a number of both formal and informal presentations to clients, users, and technical reviewers •Because of the variety of presentations and briefings that a team may be called upon to make, it is impossible to examine each of them in detail. •However, there are key elements common to most of them. •Foremost among these needs to 1. Identify the audience 2. Outline the presentation 3. Develop appropriate supporting materials 4. Practice the presentation 1. Identify the audience a team planning a briefing should consider factors such as varying levels of interest, understanding, and technical skill, as well as the available time. •Once the audience has been identified, a team can tailor its presentation to that audience. •As with other deliverables, the presentation must be properly organized and structured: •The first step is to articulate a rough outline; •The second is to formulate a detailed outline; •The third is to prepare the proper supporting materials, such as visual aids or physical models. 2. Outline the presentation •A title slide : that identifies the client(s), the project, and the design team or organization responsible for the work being presented. This slide should include company logos. •A roadmap : for the presentation that shows the audience the direction that the presentation will take. This can take the form of an outline, a flowchart, a big picture slide, and so on. •A problem statement: which includes highlights of the revised problem statement that the team produced after research and consultation with the client. •Background material on the problem: including relevant prior work and other materials developed through team research. References should be included but may be placed in a slide at the end of the presentation. •The key objectives of the client and users : as reflected in the top level or two of the objectives tree.

	•The key constraints that the design must meet. •Functions that the design must perform : focusing on basic functions, and means for achieving those functions. •Design alternatives : particularly those that were considered at the evaluation stage, including diagrams and descriptions of each. •Highlights of the evaluation procedure and outcomes: including key metrics or objectives that bear heavily on the outcome. •The selected design : explaining why this design was chosen. •Features of the design : highlighting aspects that make it superior to other alternatives and any novel or unique features. •Proof-of-concept testing : especially for an audience of technical professionals for whom this is likely to be of great interest. •A demonstration of the prototype : assuming that a prototype was developed and that it can be shown.Video or still photos may also be appropriate here. •Conclusion(s): including the identification of any future work that remains to be done, or suggested improvements to the design. 3.Develop appropriate supporting materials •At the earliest stages of the presentation planning, the design team should find out what devices (e.g., overhead projectors, computer connections, projectors, and whiteboards) are available and the general setting of the room in which it will be presenting. •Tips & pointers •Limit the number of slides. •Be sure to introduce yourself and your teammates on the title slide. •Beware of “clutter.” Slides should be used to highlight key points; they are not a direct substitute for the reasoning of the final report. •Make points clearly, directly, and simply •Use color skillfully •Use animation appropriately •Do not reproduce completed design tools •Consider carefully the size and distance of the audience 4.Practice the presentation •To be effective, speakers typically need to practice their parts in a presentation alone, then in front of others, including before an audience with at least some people who are not familiar with the topic. •we want to speak to an audience in their language, and that we want to maintain a professional tone. •Practice sessions, whether solitary or with others, should be timed and done under conditions that come as close as possible to the actual environment.
--	--

	•While practicing its presentation, a team ought to prepare for questions from its audience by: •Generating a list of questions that might arise, and their answers; •Preparing supporting materials for points that are likely to arise (e.g., backup slides that may include computer results, statistical charts, and other data that may be needed to answer anticipated questions) •Preparing to say “I don’t know,” or “We didn’t consider that.” This is very important: A team, that is, to be caught pretending to know has undermined its credibility and invited severe embarrassment.
	Module 4
	Show the design of a kid friendly craft and then depict how the design changes when considering 1) aesthetics and 2) ergonomics into consideration. Give hand sketches and explanations to justify the changes in designs
	Design 1: Basic Kid-Friendly Craft - Paper Plate Mask Design Description: Materials: Paper plate, markers, glue, construction paper, scissors, elastic cord. Steps: Cut out eye holes in the paper plate. Decorate the paper plate with markers to create a mask face. Cut out shapes from construction paper to make facial features like eyes, nose, and mouth. Glue these features onto the mask. Attach elastic cord to the sides of the mask for wearing. Now, let's consider how this design can be modified for improved aesthetics and ergonomics: Design 2: Improved Craft - Animal-Themed Mask Changes for Aesthetics:

	Color and Theme: In Design 2, we'll choose a specific animal theme, like a lion. Use vibrant colors like oranges and yellows for the mane and face. This adds a more visually appealing and thematic element for kids. Texture: Consider adding different textures to the mask, like faux fur for the mane and felt for the facial features. This tactile experience can make the craft more visually interesting and engaging. Details: Add more intricate details to the facial features, such as adding whiskers, eyelashes, and textured fur patterns. These details enhance the visual appeal. Changes for Ergonomics: Size and Shape: In Design 2, the paper plate can be trimmed to better fit the child's face. By contouring the mask to the child's face shape, it will be more comfortable to wear. Elastic Cord: Instead of a single elastic cord, use a soft, adjustable strap with Velcro fasteners. This allows for a more secure and comfortable fit without pulling or squeezing the child's head. Weight: Consider the weight of the mask. Use lightweight materials where possible to prevent strain on a child's neck and head. Hand Sketches and Explanations: For Design 2, here are hand sketches of the improved craft: Aesthetics: A sketch of the lion-themed mask with vibrant colors, faux fur, and intricate details like whiskers, eyelashes, and textured fur patterns. Ergonomics: A sketch showing how the mask is contoured to the child's face shape and the adjustable strap with Velcro fasteners for a secure and comfortable fit. These changes in design make the craft more attractive, theme-oriented, and comfortable for kids to wear, considering both aesthetics and ergonomics
--	---

Design a smart umbrella which has automatically folding and unfolding mechanism during heavy rain and sunlight with the concept of bio mimicry and sustainability. Use hand sketches to support your arguments.



Automatic

Folding and Unfolding Mechanism:

Our smart umbrella draws inspiration from nature, specifically the way certain plants and animals adapt to changing environmental conditions. To achieve automatic folding and

	unfolding, we take inspiration from the petals of a lotus flower, which repel water effortlessly and open and close in response to environmental cues. Hand Sketch 1: Automatic Folding and Unfolding Mechanism In this sketch, you can see the umbrella's frame, which is designed to mimic the lotus flower's petal-like segments. When it senses heavy rain, the segments close together, creating a waterproof canopy. In bright sunlight, the segments open up to provide shade. Biomimicry: Our smart umbrella not only incorporates the folding mechanism but also mimics the lotus leaf's self-cleaning ability. The canopy material has a hydrophobic and self-cleaning surface that repels water and debris, making it easy to maintain. Hand Sketch 2: Hydrophobic and Self-Cleaning Canopy In this sketch, you can see water droplets rolling off the umbrella's canopy, thanks to the biomimetic surface. This feature ensures that the umbrella remains clean and dry during heavy rain, promoting sustainability by reducing the need for constant cleaning and replacement. Sustainability: The smart umbrella's sustainability goes beyond its self-cleaning ability. It is designed with environmentally friendly materials and an integrated solar panel on top to harness sunlight for recharging its power source. This reduces the reliance on traditional energy sources and contributes to a more sustainable future. Hand Sketch 3: Integrated Solar Panel This sketch shows the solar panel on top of the umbrella, which captures sunlight and converts it into energy to power the automatic folding and unfolding mechanism. The energy-efficient design and sustainable materials further enhance the umbrella's eco-friendly features.
--	--

		In summary, our smart umbrella combines biomimicry and sustainability by mimicking the lotus flower's folding and self-cleaning mechanisms. The hydrophobic canopy ensures dryness during rain, and the integrated solar panel promotes renewable energy usage. This innovative design aims to make our daily lives more efficient while respecting and protecting the environment.
		Module 5
		Examine the changes in the design of a hand purse with constraints of 1) production methods 2) life span requirement 3) reliability issues and 4) environmental factors. Use hand sketches and give proper rationalization for the change in design
		Constraints of production method Since we are designing a hand purse we can have a good leather material for purse with long lasting life ● Hand purse is produced in bulk so we need large amount of production materials ● So, to reduce the cost we can use large sheets of leather material and cut the hand purse from that using machines ● The other works like stitching or combining parts of the had purse being done with workers so labours are required to do those jobs Constraints of life span requirements ● By considering the life span requirement of a hand purse it is difficult to consider the return of the product and recycling. So we must make the product with a biodegradable material. The best option for that is good quality rubber/leather. ● To get the old product back and recycle. we can give some discounts for the old product for new purchases. By giving that way we can make sure the return of the product after the use and done for the recycling purpose Constraints of reliability ● One of the main reliability issues is the long life of the hand purse in all seasons. These constraints can be resolved using the good hand purse material. ● Another constraint is breaking the straps. This can be reduced by strong materials at the end and also by stitching the joints of the purser after the instalment of the strap Constraints of the environmental factors

	● The product like hand purse the environmental factors are mainly use of the product at the time of rainy season. As we discussed before, making of the product with good quality factors can be extend the life time of the foot wear ● Due to the government. orders the product used for the making of hand purse must be biodegradable. We already mentioned that in the previous steps ● Since we are also considering the recycling steps and initiation for the products. It will help more with environmental factors. 
	Describe how to estimate the cost of a particular design of a smart bus for public use with wifi and proper cleaning facility. List the various components used. Show how economics will influence the engineering designs, use hand sketches to support the arguments.
	From the above question let us consider the cost estimation of a smart bus. Mainly for a product design there are three costs 1) labour cost , 2) material cost and 3) overhead cost. Let us discuss each ● Labour cost The labour cost of the smart bus is mainly for the designers, engineers and manufactures. ● Material cost The main material cost for smart bus is the aluminium sheets ● Overhead costs Overhead costs are the costs happening after the cost estimation. There will always be overflow of the cost estimation after the establishment and while the functioning of the smart bus. We must consider those options also.